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Overlapping construction plus a position-blind split: a curation defect in genomic sequence benchmarks and the model rankings it inverts

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Abstract

Motivation: Genomic benchmarks rarely control for near-duplicate sequences spanning their train/test split. A model can then score by recognizing memorized near-copies rather than generalizing, and no audit locates its origin.
Results: Near-duplicate leakage here requires two conditions together—an overlapping construction step and a split that ignores genomic position—rather than any property of the sequence type. A pre-registered prediction that short human regulatory tasks would be leaky failed: those GUE tasks are clean at full scale across seven independent partitions (eleven shipped tasks). The signature precedes any model: in Genomic Benchmarks the leaky positive class ships $77{,}421$ intervals on $40{,}934$ distinct coordinates, and a parameter-free coordinate screen separates seven of eight datasets. The defect is causal both ways: the omitted merge step drops the leak fraction from 0.390 to 0.012 and the random forest's gap from $+0.165$ to -0.008; imposing duplication manufactures both in a clean dataset. The benchmark then cannot rank models: the as-shipped split leaves human_enhancers_ensembl's top three of nine learners within 0.0015, yet corrected they span 23 points; a from-scratch CNN ranks fifth of ten as shipped and first corrected. Of the two leaky datasets only that one supports the ranking claim: human_nontata_promoters certifies leaky-but-correction-unsafe, its components being single-linkage chains, so its corrected delta is reported but not relied on.
Availability and implementation: Code, the report card, a near-duplicate-aware splitter and a certification tool are at `https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks`.
Contact: rikhinkavuru@gmail.com
Supplementary information: Available at Bioinformatics Advances online.

Keywords data leakage, near-duplicate sequences, benchmark evaluation, genomics, model ranking

1. Introduction

Benchmark suites are the default proving grounds for new DNA sequence classification methods, and the Genomic Benchmarks collection (Grešová et al., 2023) is used in part because it is small and quick to run. The value of any such suite rests critically on one assumption among several: that its train/test split measures generalization rather than recall of the training data. Near-duplicate leakage—exact or near-identical sequences appearing on both sides of a split—breaks that assumption, because a model can then succeed by recognizing memorized near-copies, so reported accuracy overstates generalization.

The evidence for such leakage can sit in files distributed with the benchmark, before any model is trained. The positive class of *human_enhancers_ensembl* ships $77{,}421$ genomic intervals on only $40{,}934$ distinct coordinates: $36{,}487$ coordinates appear exactly twice and $4{,}447$ once, so 94.3% of positive rows sit on a duplicated coordinate, and because the shipped split is random, 75.5% of test positives have their exact coordinate present in training. A near-default random forest (150 trees, `min_samples_leaf=1`) scores 0.860 on that split and 0.696 once the near-copies are removed; on the near-duplicate test sequences alone it scores 1.000, and on the rest 0.770—worse than the linear SVM it outranks. Four of the suite's seven binary datasets carry none of this, one is borderline under a length-robust metric, and two are leaky, so the defect is a property of particular datasets rather than of the suite.

Leakage is an established benchmarking hazard. Kapoor and Narayanan (2023) document it as a pervasive driver of the reproducibility crisis in ML-based science, inflating performance and, in extreme cases, changing the apparent ordering of methods. In genomics, hashFrag (Rafi et al., 2026) shows homologous sequences spanning cis-regulatory splits inflating neural-model performance, and reports that models rely on memorized associations wherever function is conserved between homologs; standard protein-interaction splits leave most test cases with near-duplicate counterparts (Bushuiev et al., 2024); de-leaked test sets sharply reduce reported performance including for AlphaFold-Multimer (Kovtun et al., 2024); and on LIT-PCBA a trivial memorization baseline matches sophisticated methods because the benchmark is leaky

(Huang et al., 2025). Evaluation design is itself a recognized genomics hazard (Whalen et al., 2022; Schreiber et al., 2020; Toneyan et al., 2022). Outside biology the ranking question has been posed most sharply: Barz and Denzler (2020) find near-duplicates of training images inside the CIFAR test sets, Northcutt et al. (2021) show test-set defects destabilizing the ranking of models evaluated on ten widely used benchmarks, and under controlled injection Zhou et al. (2023a) show benchmark leakage distorting model comparison, a hazard established at scale by work on train/test overlap, corpus deduplication and language-model contamination (Elangovan et al., 2021; Lee et al., 2022; Sainz et al., 2023; Magar and Schwartz, 2022; Deng et al., 2024). The usual remedy is a similarity-aware partition (Hobohm et al., 1992; Walsh et al., 2016; Teufel et al., 2023; Petti and Eddy, 2022; Li and Godzik, 2006; Steinegger and Söding, 2017).

What that literature lacks, and a small curated suite supplies, is a matched clean control inside the same collection and a construction step that can be edited. The open question is accordingly upstream of the remedy: not how to repair a leaky split, but where the leakage originates and whether it reorders which model wins. Two candidate answers are separable. Leakage may track what a dataset contains—sequence type, task, window length—in which case a practitioner can screen by task family. Or it may track how the dataset was assembled, in which case the screen belongs in coordinate space and the fix belongs in the curation pipeline. Distinguishing them requires a prediction committed in advance and an intervention on the construction step alone. Pretrained foundation models are outside this evidence: DNABERT-2 (Zhou et al., 2023b), HyenaDNA (Nguyen et al., 2023) and the Nucleotide Transformer (Dalla-Torre et al., 2024) all pretrain on the GRCh38 human reference—HyenaDNA on it alone, the other two as the human component of a multispecies corpus—and these test sequences lie in it (§4), so evaluating them would open a second, pretraining-based leakage channel no split can close.

Across the datasets audited here, leakage tracks how a dataset was assembled rather than what kind of sequence it contains, and the condition it requires is conjunctive: an overlapping construction step *and* a split that ignores genomic position (§4). Where similarity is instead a property of the sequences themselves—GUE's *virus_covid*, whose nine SARS-CoV-2 variants differ by a handful of mutations—a leak fraction of 1.000 is biology and carries no information about curation (§4). We measure a coordinate-space signature on the shipped genomic intervals rather than the sequences, and a screen with no fitted parameter separates the suite $7/8$ and calls four out-of-suite BEND partitions clean; we edit the construction step in both directions, applying the omitted merge step to a leaky dataset to abolish the leakage and its downstream consequences, and imposing it on a clean dataset to manufacture both; and we score a prediction committed before the target suite was examined, that GUE's short human regulatory tasks would be leaky, which failed—all eleven are clean at full scale, refuting the task-type hypothesis the prediction encoded. The downstream cost is a benchmark that cannot rank models, demonstrated on *human_enhancers_ensembl*, where the as-shipped split cannot separate its top three of nine learners—within 0.0015—yet those three span 23 accuracy points corrected. That demonstration requires a memorization-prone model in the comparison set and a split whose construction leaves near-duplicates spanning it; both datasets on which it holds are short human regulatory sets, one fixed at 251 bp and one variable at a 269 bp median over a 2–573 range, so that regime remains a possible further necessary condition, and is not a sufficient one. We release a per-dataset report card, an aligner-free near-duplicate-aware splitter, and a certification tool returning *leaky-but-correction-unsafe* where whole-cluster correction is not safe.

2. Methods

2.1. Datasets

We study the seven binary Genomic Benchmarks datasets (Grešová et al., 2023) and additionally report the three-class *human_ensembl_regulatory* set; sizes, lengths and balance are in Supplementary Table S1. Their upstream provenance is pinned rather than assumed: per the curators' own construction notebooks, *human_nontata_promoters* and *human_enhancers_cohn* derive from the Ensembl release-97 GRCh38 toplevel DNA FASTA, and *human_enhancers_ensembl* and *human_ocr_ensembl* from the release-100 Regulatory Build external-feature table over the release-100 GRCh38 primary assembly. Every upstream URL was re-resolved on 2026-07-20 (all HTTP 200; `results/reference_provenance.csv`), and archive hosts are recorded alongside the live ones—`https://jul2019.archive.ensembl.org` for release 97 and `https://apr2020.archive.ensembl.org` for release 100—because Ensembl release 116 (June 2026) is the last served on the current platform, with data moving to a new site from summer 2026, so the live paths in the notebooks are not durable citations. Balanced datasets are curated to 0.500 positive fraction in both train and test, so class balance is preprocessed, not natural. All censuses and leak fractions are full-scale; four accuracy deltas are not. The corrected-split deltas for *human_ocr_ensembl*, *demo_human_or_worm* and *human_enhancers_cohn*, and the clean-control roster arm of §3.6, run on $20{,}000$-sequence subsamples of datasets whose report-card column is headed n (full). Their full-scale leak fractions are ≤ 0.012 (Jaccard) and ≤ 0.068 (containment), and subsampling can only *hide* leakage, so a subsample is admissible there to confirm cleanliness—never to establish it.

2.2. Coordinate-space construction signatures

Genomic Benchmarks distributes per-sequence intervals (`id,region,start,end,strand`) separately from the sequences its loader returns. We recover them offline and realign them to loader row order, which is deterministic (split, then sorted class directory, then sorted filename, the stem being the interval `id`); recovered interval width equals sequence length for 100% of rows in all eight datasets, a check informative only for the four variable-length ones. These coordinates are read on a separate code path from the k-mer clustering and are causally upstream of it, so they give non-circular corroboration rather than a restatement. Per class we compute self-overlap redundancy $R_{\mathrm{self}} = 1 - n_{\mathrm{merged}}/n_{\mathrm{raw}}$ and the share of *test* intervals overlapping any *train* interval at $\geq t$ reciprocal overlap, $t \in \{\geq 1\,\mathrm{bp}, 0.5, 0.9, 1.0\}$; overlap is strand-agnostic, strand being unrecorded for the two *demo* datasets. The screen thresholds the maximum over classes of the $\geq 50\%$ statistic at 0.1—the cut already used in sequence space—so it adds no fitted parameter.

The assembly is measured, not assumed, and the unsampled check is the stronger one. All $190{,}973$ shipped intervals of the two leaky datasets name a primary chromosome and none an alternate contig, and—without sampling, so with no denominator to choose—915 of them (0.48%) end past a GRCh37 chromosome boundary while none ends past a GRCh38 one. A sampled sequence-level check agrees. For 49 test intervals stratified over both datasets, classes and strands we fetched the reference at the shipped coordinates from GRCh38 and GRCh37 via the UCSC REST API on 2026-07-20—`hg38`, the Dec. 2013 assembly, accession GCA_000001405.15, and `hg19`, the Feb. 2009 assembly, accession GCA_000001405.1—reading intervals as 0-based half-open and reverse-complementing minus-strand records: GRCh38 reproduces the shipped sequence in 49 of 49 cases and a 1-based reading in none of 49. The GRCh37 denominator is 48, not 49, for the same reason the boundary count gives: one sampled chrX interval runs past the GRCh37 chromosome end, so the API returns an error rather than a sequence and that row is unevaluable on GRCh37 by construction. On the 48 evaluable rows GRCh37 reproduces the shipped sequence once.

2.3. Construct-and-break manipulations

We intervene on the construction step alone. *Fix-a-leaky*: *human_enhancers_ensembl* receives the merge step its curators omitted—every duplicated coordinate collapsed to one representative row—with sequences, labels, model code and split ratio unchanged and negatives subsampled to restore the control's balance. *Break-a-clean*: *human_ocr_ensembl*, drawn from the merged Ensembl Regulatory Build and clean by every measure, is rebuilt as an un-deduplicated assembly, duplicating positives until the same 94.3% share of positive rows sits on a multiplicity-2 coordinate as on *human_enhancers_ensembl* (requiring 89.1% of them, since duplicating a fraction r leaves $2r/(1+r)$ of rows on a duplicated coordinate), then downsampling to the control's exact size and class balance. Each manipulation is paired with an unmanipulated control at the same split seed. In both manipulated datasets the duplicated coordinates sit in the positive class, so both interventions perturb balance and size unless corrected; balance-matched arms are primary, and the break-a-clean pair is matched on size as well while the fix-a-leaky pair cannot be, since size there cannot be equalised without discarding the rows the intervention is defined to remove. A ten-dose version sweeps the imposed duplication rate (Supplementary §S2).

2.4. Near-duplicate measurement and re-split

We measure train/test overlap by the exact Jaccard similarity of k-mer sets ($k = 8$) over all test–train pairs via sparse binary matrix products; the leakage fraction is the share of test sequences whose maximum similarity to any training sequence exceeds a threshold (0.5, 0.7, 0.9). A k-mer Jaccard is bounded by the length ratio, so it undercounts leakage on datasets with disparate lengths—across *human_enhancers_ensembl*'s test–train pairs, 61% have their Jaccard held below 0.7 by length ratio alone—so we also report the length-robust *containment* index $|A \cap B| / \min(|A|, |B|)$. Jaccard leak fractions are lower bounds; unfloored containment runs the other way and is an *upper* bound on variable-length data, since a very short sequence is contained in almost anything. Calibrated on synthetic sequences with known ground truth, the 8-mer Jaccard at 0.7 detects an exact copy, one point mutation and a window shifted by up to 10% of its length at every length we use, but at $70\,\mathrm{bp}$ it misses two point mutations 70% of the time where at $\geq 300\,\mathrm{bp}$ it tolerates five: a near-exact-duplicate detector on short sequences, a homology detector on long ones, so threshold choice should follow length (Supplementary §S4). We separately census exact byte-identical duplicates on a code path using no k-mers: *human_enhancers_ensembl* has $11{,}774$ such test sequences (0.380 of its test set), whereas *human_nontata_promoters* has 0 exact duplicates yet 0.225 near-duplicates at Jaccard ≥ 0.9—the more homology-like but statistically weaker case.

To correct leakage we connect sequences whose Jaccard exceeds 0.7, take connected components as clusters, and assign whole clusters to train or test, preserving the original ratio and balance and guaranteeing zero residual cross-split similarity above the threshold. Sweeping the threshold, the drop is flat on *human_enhancers_ensembl* ($0.156/0.156/0.157$ at $0.5/0.7/0.9$; 0.158 under largest-component removal) but scales strongly on *human_nontata_promoters* ($0.205/0.121/0.006$), since 0.9 leaves its large $[0.5, 0.9)$ band uncorrected. This is an aligner-free instance of GraphPart's whole-cluster partitioning (Teufel et al., 2023); an MMseqs2 arm is in Supplementary §S3.

2.5. Models, decision rule and pre-registration

Sequences are represented by overlapping k-mer counts, $k \in \{4, 6\}$. Nine classifiers span *memorization propensity* (equivalently, effective regularization) rather than raw capacity: logistic regression, a linear SVM (`LinearSVC`), a random forest (150 trees at scikit-learn defaults, `min_samples_leaf=1`), `HistGradientBoosting`, extremely randomized trees, 1- and 15-nearest neighbours, a multilayer perceptron (MLP) and Gaussian naive Bayes; the first four constitute the four-model comparison, all nine the roster. Scale-sensitive learners receive L2 normalization, tree ensembles raw counts. All are trained from scratch on each split, so none carries a pretraining confound. Accuracies use `.predict()` (`argmax`); the headline drop is the original accuracy minus the mean corrected accuracy over five re-split seeds. Parameters and roster rationale: Supplementary §S1. An inversion criterion built for four models is too permissive for nine—"the top model changed by more than a point" fires on a *clean* control from repartitioning alone—so we additionally require that the two models exchanging the top slot each lead on their own split by a *paired* cluster-bootstrap difference excluding zero, and we report the swap margin separately from the as-shipped first-to-second margin, which can be a tie when the former is large.

The deep-model probe is a from-scratch 1D residual convolutional network in the Basset (Kelley et al., 2016) / DeepSEA (Zhou and Troyanskaya, 2015) lineage (one-hot input, no k-mer features), trained on each dataset's training split only. We pre-specified, in a document committed alongside the results it predicts, the narrowed claim, a dropout×weight-decay grid, and a refutation condition: if the standard-practice regularized cell (dropout 0.3, weight decay 10^{-4}) has a cluster-bootstrap drop CI including 0 on *both* datasets, the strong "any expressive learner is inflated" reading is refuted and reported as a limitation. This is a binding stated condition, not an externally timestamped registration; it was committed in the same commit as the experiment code and so carries no priority evidence at all, unlike the cross-suite registration below. Commit hashes and timestamps for both, with the `git log` output that establishes the ordering and an explicit statement of what author-controlled timestamps can and cannot show, are published as `PREREGISTRATION.md` in the code repository.

Six predictions were committed in code, inside the module that would test each, before that module was run, and all are scored in the Results whether or not they held: P1 1-nearest-neighbour shows the largest graded memorization gap, registered against the *raw* form of that statistic—the prevalence-corrected form is exploratory here and is registered forward for the next dataset, not counted backward (§3.7); P2 it shows the largest split-induced drop; P3 the four memorization-prone learners (1- and 15-nearest neighbours, the forest, extremely randomized trees) outrank the two linear models as shipped and fall below them after correction; P4 on a clean control no model's drop interval excludes zero; P5 cluster-grouped cross-validation selects a more regularized forest than naive cross-validation; P6 a ranking inversion occurs at every imposed leak fraction above the diagnostic threshold of Supplementary §S2 and at none below. Separately, before that suite was examined, we registered for GUE (Zhou et al., 2023b) that its short fixed-length human regulatory tasks would be leaky and its multi-species tasks clean (§3.12).

The *graded memorization gap* is $\mathrm{acc}[\geq 0.9] - \mathrm{acc}[< 0.5]$. Raw, it is confounded: the strata differ sharply in class composition, in opposite directions on the two leaky datasets, and a constant classifier that memorizes nothing scores $+0.803$ on *human_enhancers_ensembl* and -0.980 on *human_nontata_promoters*. We therefore report the gap from *balanced* accuracy within each stratum, for which a constant classifier scores exactly zero, alongside the raw gap so the confound stays visible (Supplementary §S1).

2.6. Controls, cross-suite census and intervals

Chromosome holdout and regularization.
Using the recovered coordinates we assign whole chromosomes to the test side in a single greedy pass until the target test fraction is reached (nine chromosomes for *human_enhancers_ensembl*, twelve for *human_nontata_promoters*; there is no cross-validation loop). This field-standard control is deployment-relevant but *not* leakage-free: residual leakage at Jaccard 0.7 is 0.8% and 0.02%, against the exact 0 the near-duplicate-aware split guarantees. Separately we sweep, for the random forest only, `min_samples_leaf`$\in \{1, 5, 20, 50, 100, 200, 500\}$ and `max_depth`$\in \{\text{None}, 16, 8, 4\}$ at full scale—the largest settings exceed the largest cluster, so no leaf can be a pure duplicate block—and cross-validate the same leaf grid on the as-shipped *training* set under ordinary stratified and cluster-grouped 5-fold.

Cross-suite census.
The census is applied unchanged to the Nucleotide Transformer downstream tasks (Dalla-Torre et al., 2024) and to GUE (Zhou et al., 2023b); the full pipeline is then run on the tasks the first flags, with a clean task as control. Collapsing two nestings leaves eleven independent tasks of the thirteen shipped, and we report independent-task counts throughout; the three splice tasks are not nested and count separately, and we draw no before-and-after inference between the original and revised releases, which are not the same data recurated. One deviation affects the anchoring task: *enhancers* ships only 400 test sequences, too few for a usable interval, so for that task alone we pool train and test and draw a fresh stratified $80/20$ split before running both arms. Pooling redistributes the duplicate pairs, so the re-split leak fraction ($\varphi = 0.099$) is not the same quantity as the 25% byte-identical figure the as-shipped split carries (Supplementary §S1).

Injected multiclass construction.
The only multiclass set is clean, so we answer the question by construction, injecting label-carrying near-duplicate train→test copies. Here f is the fraction of *training* rows copied and appended to the test set, so realised test-set leak fractions are $\varphi = 0.29$, 0.44 and 0.62 at $f = 0.1$, 0.2, 0.4, not f. The arm runs on a $20{,}000$-sequence subsample; the dose is imposed rather than measured, so scale is not load-bearing.

Intervals and reproducibility.
Because near-duplicates violate the independence a sample-wise bootstrap assumes, we resample whole within-test near-duplicate components as blocks on *both* arms, quantify the design effect using Kish's effective block size rather than the mean (these distributions are heavy-tailed and the mean-size convention understates the factor roughly twofold), report it beside the cluster-versus-sample variance ratio the bootstrap measures directly, cross-check against a Liang–Zeger cluster-robust standard error, and fold the five re-split-seed variance into a combined-source interval; Benjamini–Hochberg at $q = 0.05$ is applied over a declared family of 15 deltas (Benjamini and Hochberg, 1995). Reported 95% CIs are percentile bootstrap intervals over $10{,}000$ draws with the resampling unit stated per number; the three combined-source intervals are instead Wald, $\delta \pm z\sqrt{s^2_{\mathrm{boot}} + s^2_{\mathrm{seed}}}$, and labelled where they appear. Models are not refit while the test set is resampled, so every interval is conditional on the fitted models. The classical pipeline is deterministic, seeded and CPU-only; the CNN is not, since 12 of 20 committed grid cells carry dropout > 0 and dropout masks are drawn from the device RNG, so CNN results are device-dependent within the five-seed tolerance we report. Because `genomic_benchmarks` $1.0.0$ re-extracts on every call we cache locally and pin the cached bytes by SHA-256, recording row counts, extraction timestamps and both file and content digests for all nine cached combinations (Supplementary §S1.8); audits

Table 1. Near-duplicate leakage report card. **Verdict** is LEAKY if the full-scale near-duplicate test fraction exceeds 0.1 at Jaccard 0.7; a "clean" verdict means only *no forward-strand near-duplicate leakage detected*, subject to the disclosed length-pair cap—it is not a certificate of homology-freedom. "leak@0.7 (Jac / con)" gives the leak fraction under the 8-mer Jaccard and the length-robust containment index; "Acc. lost" is the best-model accuracy change under the near-duplicate-aware re-split (positive = accuracy lost). Leak fractions and censuses are full-scale on every row, but four of these deltas—*human_ocr_ensembl*, *demo_human_or_worm*, *human_enhancers_cohn* and the three-class *human_ensembl_regulatory*—are computed on $20{,}000$-sequence subsamples of the datasets whose n column is headed *full*. Subsampling can only hide leakage, never create it, so a subsampled delta is admissible here to *confirm* cleanliness and never to establish it (§2.1). *demo_coding_vs_intergenomic_seqs* is *not* in that group: its delta is full-scale at $n = 100{,}000$. †*demo_coding_vs_intergenomic_seqs* is clean under Jaccard but borderline under containment. ‡*leaky-but-correction-unsafe*: the verdict the certification tool returns for a LEAKY dataset whose largest-cluster cohesion falls below 0.5 (*human_nontata_promoters*, 0.084). Its near-duplicate components are single-linkage chains rather than sets of mutual near-copies, so whole-cluster re-splitting quarantines related but non-duplicate sequences and removes genuine signal along with the leakage. The leakage verdict stands; it is the *corrected* delta on that row that cannot be read as a clean measurement, and the novel-stratum accuracy on the as-shipped split should be reported in its place.

Dataset	n (full)	leak@0.7 (Jac)	leak@0.7 (con)	Verdict	Acc. lost	RF rank	Reorders under chrom.-holdout?
human_nontata_promoters	36,131	0.406	0.445	**LEAKY**‡	$+0.118$	$1 \to 3$	no (RF stays rank 1)
human_enhancers_ensembl	154,842	0.384	0.845	**LEAKY**	$+0.165$	$1 \to 4$	yes (RF → rank 4)
demo_coding_vs_intergenomic_seqs	100,000	0.078	**0.131**	borderline†	-0.001	$4 \to 4$	n/a
human_ocr_ensembl	174,756	0.010	0.068	clean	$+0.004$	$4 \to 4$	n/a
demo_human_or_worm	100,000	0.012	0.042	clean	-0.004	$4 \to 4$	n/a
drosophila_enhancers_stark	6,914	0.016	0.033	clean	$+0.012$	$2 \to 3$	n/a
human_enhancers_cohn	27,791	0.001	0.005	clean	-0.004	$4 \to 4$	n/a
human_ensembl_regulatory (3-class)	289,061	0.005	–	clean	-0.010	n/a	n/a

against a different release should re-derive the report card and will see the content digests move.

3. Results

3.1. Leakage is dataset-specific, and one clean verdict moved

Only *human_enhancers_ensembl* and *human_nontata_promoters* exceed a 10% near-duplicate test fraction at Jaccard 0.7 (0.384, 0.406); the three-class set is clean at 0.005 (Table 1). Re-measuring at full scale with containment reveals substantially more leakage on *human_enhancers_ensembl* ($0.384 \to 0.845$) and flags one dataset the length-blind metric misses: *demo_coding_vs_intergenomic_seqs*, clean under Jaccard (0.078), crosses the cut under containment (0.131), so we relabel it *borderline*. The remaining four stay well below under both metrics (largest containment 0.068). Two datasets are leaky—one of them, *human_nontata_promoters*, certified *leaky-but-correction-unsafe* because its largest-cluster cohesion is 0.084, so whole-cluster correction would strip genuine signal along with the leakage—one borderline, four clean. A clean verdict moved, so both metrics should be reported.

3.2. The cause is a curation defect visible in the coordinates

Of the $77{,}421$ positive intervals in *human_enhancers_ensembl*, only $40{,}934$ are distinct: $36{,}487$ coordinates ship exactly twice and $4{,}447$ once, so 94.3% of positive rows sit on a multiplicity-2 coordinate (Fig. 1A). That coordinate redundancy is the sequence redundancy to a close approximation, and the two counts coincide to 0.7%. In coordinate space, $11{,}694$ test positives ship a coordinate that also occurs among the training positives—a direct count on the shipped split, and 0.15% above the $11{,}676$ a class-stratified uniform split predicts, $36{,}487 \times 2 \times \frac{15{,}485}{77{,}421} \times \frac{61{,}936}{77{,}420}$, one duplicated coordinate contributing such a row exactly when one of its two rows lands in test. In sequence space, on a code path using no coordinates, $11{,}774$ test sequences are byte-identical to a training sequence. The agreement is between two independent counts; per-pair identity was not checked on all $36{,}487$ pairs.

Thresholding the maximum-over-classes cross-split coordinate overlap at the same 0.1 cut used in sequence space separates the suite $7/8$, the two leaky datasets reading 0.755 and 0.992 against 0.073 and below for the rest (Fig. 1B). The single miss is *demo_coding_vs_intergenomic_seqs*, containment-borderline but coordinate-clean: its leakage is coding-paralog homology and its `coding_seqs` class is keyed by transcript accession, one region per row, so its coordinate statistics are structurally zero rather than measured; coordinates may escalate a verdict and never clear one. Two refinements were forced by the data: redundancy is not always in the positive class, *human_nontata_promoters* carrying essentially all of its own in the *negative* class ($R_{\mathrm{self}} = 0.961$ against 0.047); and mere adjacency over-calls, *drosophila_enhancers_stark* tiling at a 2142 bp median with base-pair-scale touches, 0.406 at ≥ 1 bp but 0.036 at $\geq 50\%$ reciprocal overlap.

A comparison internal to Ensembl ties the defect to a missing deduplication step: *human_ocr_ensembl* and *human_ensembl_regulatory*, from the release-100 Regulatory Build's merged "MultiCell" segmentation (Zerbino et al., 2015), sit at 0.000–0.010 self-overlap where the un-deduplicated *human_enhancers_ensembl* sits at 0.471. That is an association, not a cause—the upstream resources differ, so it is *not* one pipeline run with and without a merge step. Classifying all eight datasets by documented positive-class construction reproduces

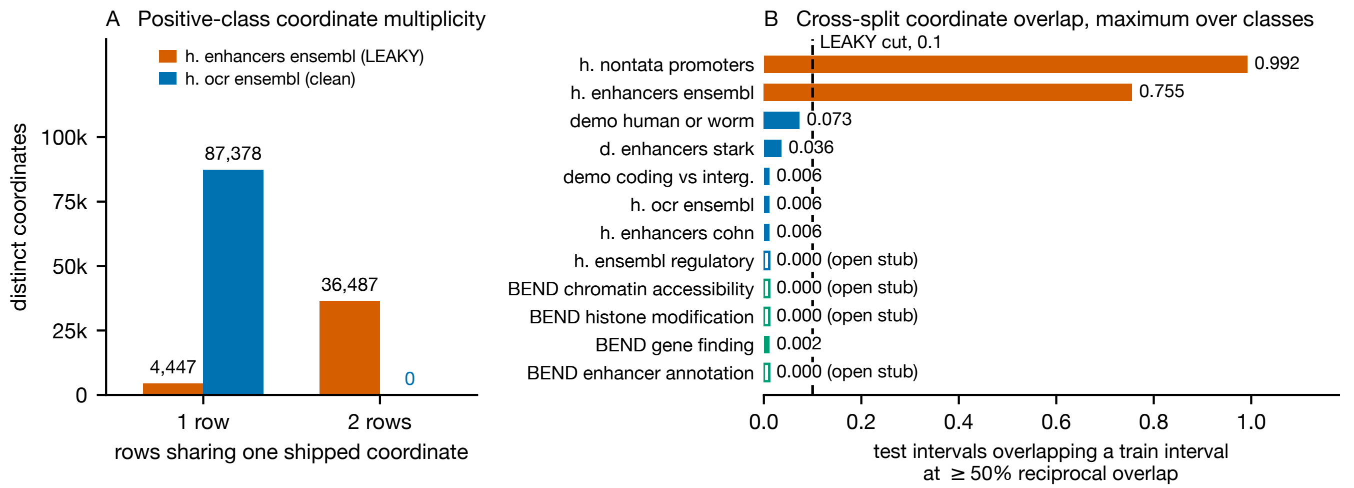

Figure 1 The construction defect in the shipped coordinates, measured before any model is trained. (A) Distinct positive-class coordinates by the number of rows sharing them, for the leaky *human_enhancers_ensembl* and the clean *human_ocr_ensembl*; $36{,}487$ of the former's coordinates carry two rows each, 94.3% of its positive rows. (B) Fraction of test intervals overlapping any training interval at $\geq 50\%$ reciprocal overlap, taken as the maximum over classes, for the eight Genomic Benchmarks datasets and the four BEND partitions (green); the dashed rule at 0.1 is the LEAKY cut used throughout, carried over unchanged from sequence space, so the screen adds no fitted parameter. A measured 0.000 is drawn as an *open* hairline stub and labelled as such, so that it is distinguishable both from a small non-zero bar and from a missing one; four partitions read exactly zero. Overlap is strand-agnostic. *demo_coding_vs_intergenomic_seqs* is the one miss: it is borderline in sequence space under containment, but one of its classes is keyed by transcript accession rather than by interval, so its 0.006 is structurally zero rather than measured.

Table 2. Positive-class construction step against measured verdict. **Retrospective**: categories were assigned with the verdicts already known, so the table has zero free parameters but is not a prospective test. The prospective coordinate call is BEND (§3.4); the prospective sequence-space call is GUE (§3.12), which the prediction got wrong. ‡borderline under containment only.

Dataset	Construction of positives	Merge step	Verdict
h. enhancers_ensembl	un-dedup. assembly	none	LEAKY
h. nontata_promoters	fixed window, recurring loci	none	LEAKY
h. ocr_ensembl	merged segmentation	MultiCell	clean
h. ensembl_regulatory	merged segmentation	MultiCell	clean
h. enhancers_cohn	curated, deduplicated	curation	clean
d. enhancers_stark	curated, deduplicated	curation	clean
demo_coding_vs_interg.	random genome draw	disjoint draw	clean‡
demo_human_or_worm	random genome draw	disjoint draw	clean

Table 3. Construct-and-break manipulations. Each intervention is paired with an unmanipulated control run of the identical pipeline at the same split seed and matched class balance; the break-a-clean pair is matched on size as well. Control rows are fresh seed-0 random splits, not the shipped partition. "Leak" is the near-duplicate test fraction at Jaccard 0.7; "Rank" is the random forest before→after correction, of four models. Intervals are two-arm cluster bootstraps over whole within-test components; * marks intervals excluding zero, and control and intervention intervals are disjoint in both directions.

	Condition	n	Leak	RF drop [95% CI]	Rank
Fix	*ensembl* control	154,842	0.390	$+0.165$ $[0.157, 0.173]^*$	$1 \to 4$
	dup. collapsed	81,868	0.012	-0.008 $[-0.019, 0.003]$	$4 \to 4$
Break	*ocr* control	20,000	0.004	$+0.002$ $[-0.019, 0.025]$	$4 \to 4$
	pos. duplicated	20,000	0.204	$+0.105$ $[0.083, 0.126]^*$	$1 \to 4$

every verdict (Table 2), but retrospectively. The causal claim rests on §3.3.

3.3. Editing the construction step alone creates and cures the reordering

Editing the construction step alone abolishes the syndrome in a leaky dataset and manufactures it in a clean one, turning the correlational coordinate evidence into a causal claim in both directions (Table 3). Applying the omitted merge step to *human_enhancers_ensembl*, at matched balance, collapses its leak fraction from 0.390 to 0.012 and the random-forest inflation from $+0.165$ $[0.157, 0.173]$ to -0.008 $[-0.019, 0.003]$—a null—and the reordering disappears, the forest sitting at rank 4 before and after correction. Imposing the same defect on the merged, clean *human_ocr_ensembl*, at identical size *and* balance, raises its leak fraction from 0.004 to 0.204 and its inflation from $+0.002$ $[-0.019, 0.025]$ to $+0.105$ $[0.083, 0.126]$, and manufactures the reordering: the forest is promoted to rank 1 on the contaminated split and demoted to rank 4 again by correction—the syndrome of §3.6, produced on demand. The effect is larger under matching than without it ($+0.063 \to +0.105$), so the confounds were suppressing rather than creating it.

Duplicated rows scattered across a random split are leaky by construction; what the manipulation adds is that the full downstream syndrome, including the memorizer's promotion to rank 1, follows from the construction step alone. Size matching removes some of the duplicate partners it has just

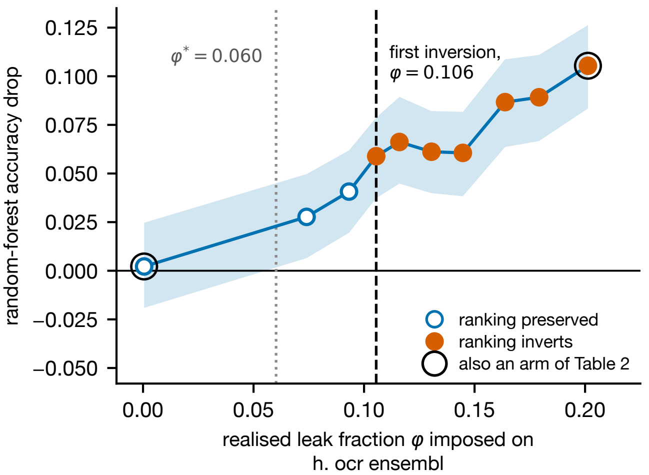

Figure 2 Ten doses of the same construction defect, imposed on the clean *human_ocr_ensembl*. Random-forest accuracy drop under near-duplicate-aware correction against the realised test-set leak fraction φ at Jaccard 0.9; the shaded band is the two-arm cluster-bootstrap 95% CI over whole within-test near-duplicate components, and each dose is a single fit at one split seed. Filled points are doses at which the four-model ranking inverts, open points doses at which it does not. Circled points are the two arms of Table 3 re-reported rather than fresh trials, so the sweep supplies eight additional doses. The dashed rule marks the first inversion, the dotted rule the diagnostic threshold φ^* computed at that dose.

introduced, so the imposed leak fraction (0.204) is about half the undownsampled value (0.498) and the estimate is conservative. A ten-dose sweep on the same clean dataset places the first ranking inversion at an imposed leak fraction of 0.106 and finds none below it, across eight doses beyond the two arms of Table 3 (Fig. 2; Supplementary §S2). Editing one curation step therefore both removes and installs the reordering, at a single fit per condition read only between arms.

3.4. An out-of-suite coordinate census

Because the screen needs only shipped coordinates it runs outside the suite, at lengths far beyond anything audited here in sequence space. Applied unchanged to the four supervised tasks of BEND (Marin et al., 2024), whose BED releases carry their own split assignments, it returns 0.000 at $\geq 50\%$ reciprocal overlap on chromatin accessibility ($1{,}410{,}554$ train against $372{,}153$ test intervals), 0.000 on histone modification ($433{,}861$ against $120{,}567$) and 0.000 on enhancer annotation in all ten folds—the last at $100{,}096$ bp per sample (Fig. 1B). Gene finding, the one task partitioned at 80% identity rather than by chromosome, reproduces the *drosophila_enhancers_stark* pattern: 0.183 of its 597 test intervals touch a training interval at ≥ 1 bp, yet only 0.002 reach $\geq 50\%$ and none 90%. This is the only prospective construction-based call in the paper, and it is near-tautological on the three chromosome-split partitions. Together with the $7/8$ in-suite separation, the screen therefore reads construction defects across four orders of magnitude of interval length without being refitted to the suite it was developed on.

3.5. A near-duplicate-aware re-split lowers accuracy only on leaky datasets

Under the re-split the best model's accuracy drops sharply on the leaky datasets and nowhere else: 16.5 points on *human_enhancers_ensembl* $(0.8596 \rightarrow 0.6951 \pm 0.0019$ over five re-split seeds; combined-source Wald CI $[15.4, 17.4])$ and 11.8 on *human_nontata_promoters* $(0.9284 \rightarrow 0.8101 \pm 0.0055)$. For that second dataset point estimate and interval are computed on different estimands. The five-seed mean gives the 11.8-point figure; the combined-source interval—which additionally folds in the five re-split-seed variance, exactly as the *human_enhancers_ensembl* figure above does—is computed on the seed-0 partition, where the delta is 10.8 points, giving $[8.8, 12.9]$; the cluster bootstrap alone gives $[9.2, 12.6]$. Every clean-dataset delta includes zero for the memorizing forest as well as the linear models, the largest being *drosophila_enhancers_stark* at $+0.024$, cluster-bootstrap CI $[-0.007, 0.054]$. The negative control confirms causation: a random re-split at matched size and balance leaves the leaky-dataset accuracies unchanged (best-model deltas -0.001 and -0.002, point estimates, two orders of magnitude below 0.118 and 0.165; Supplementary Fig. S1), so only removing near-duplicates, not re-splitting per se, lowers the score. Corrected accuracy is stable across five re-split seeds and, independently, five forest training seeds $(0.698 \pm 0.002$; $0.821 \pm 0.003)$, the forest demoted on every seed. That de-leaking lowers apparent accuracy is established across domains (Kapoor and Narayanan, 2023; Rafi et al., 2026; Bushuiev et al., 2024); what the within-suite version adds is the matched clean control, on which the same procedure moves nothing.

3.6. The consequence: the benchmark cannot rank models

The four-model comparison, and where the fall comes from.
On *human_enhancers_ensembl* the best model *changes*. In the four-model comparison the near-default random forest, ranked first as shipped (0.860), falls to last after correction (0.696) and *LinearSVC* becomes best $(0.798$; Fig. 3). This holds under accuracy, AUROC (0.805 against 0.870) and F_1 (0.635 against 0.796), so it is not a threshold artifact; the bootstrap probability that the forest ranks first goes from 1.0 to 0.0; and it reproduces under the chromosome-holdout control (§3.13). The fall decomposes, and only one part is the split: on the as-shipped split's *novel* stratum, with no re-splitting and no retraining, the forest scores 0.770 against *LinearSVC*'s 0.782—the ranking has already changed, but by only 1.2 points. The remaining 10 points is a *retraining penalty*, de-leaking removing near-duplicate partners from the training set too, which costs the memorizer more than the others (residual -0.074 against $+0.012$ to $+0.016$).

Nine learners: a tie as shipped, 23 points corrected.
Widening from four learners to nine sharpens the statement (Supplementary Table S2). The leaky benchmark cannot separate its top three at all: the MLP (0.8736), 1-nearest-neighbour (0.8729) and extremely randomized trees (0.8721)

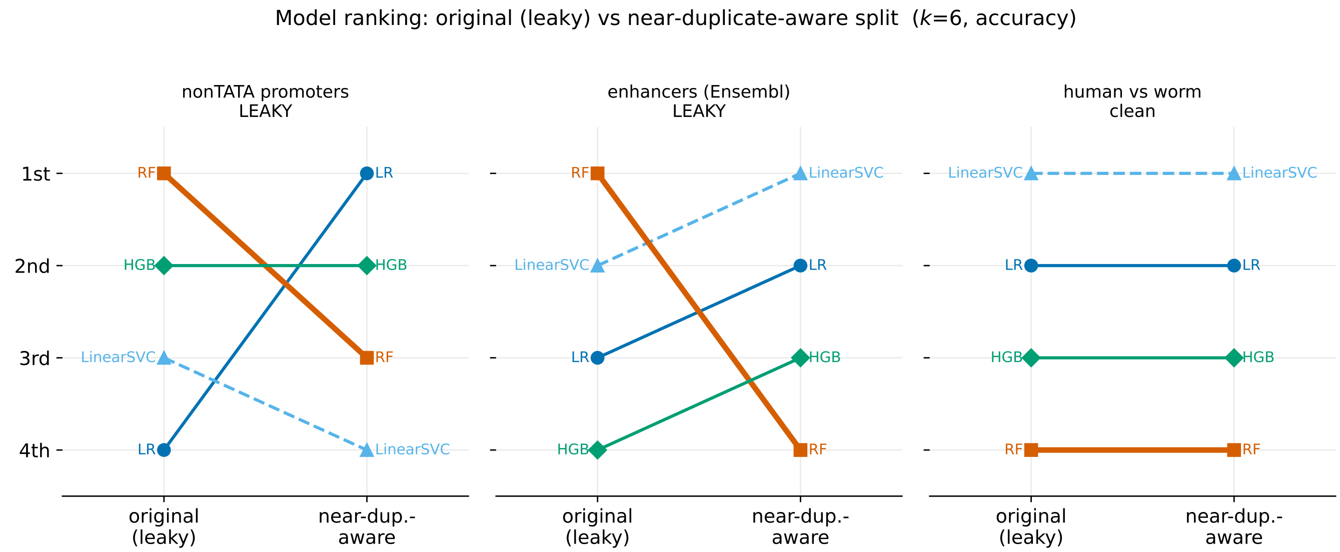

Figure 3 Four-model rankings on the original (leaky) split versus the near-duplicate-aware split (k=6, accuracy). The random forest is dethroned on *human_enhancers_ensembl* (middle) and, under accuracy only, on *human_nontata_promoters* (left), while rankings are stable on the clean control *demo_human_or_worm* (right, 20k subsample). Kendall τ is deliberately not annotated: at four models it is too coarse to carry (§3.6), and it orders these panels opposite to the conclusions they support.

finish within 0.0015, and the paired interval on the top-two margin, $[-0.005, 0.006]$, includes zero. On the corrected split those three span 23 accuracy points (0.768, 0.537, 0.628). The benchmark is not merely crowning the wrong model; it is blind to enormous real differences between models it ranks as equals. The linear SVM rises from rank 5 to 1 and logistic regression from 6 to 2, while 1-nearest-neighbour falls from rank 2 to 9, last, at 0.537—3.7 points above the 0.500 base rate of this balanced test set, on $n = 30{,}971$. The two models exchanging the top slot each lead on their own split by a paired margin excluding zero ($[0.071, 0.080]$ as shipped, $[0.024, 0.035]$ corrected). The four models whose drops exclude zero are 1-NN, the forest, extremely randomized trees and the MLP; the other five—both linear models, boosting, naive Bayes and 15-nearest-neighbour, which we designated memorization-prone in advance—are null. The MLP is not one of those four designated learners, so its presence says the inflation reaches a neural learner too.

The convolutional network.

The leaky split ranks it fifth of ten; the corrected split ranks it first. The registered nine are unaffected by adding it—the linear SVM remains the corrected winner *of the registered nine*—and the CNN is a tenth, *post hoc* entrant. Replicating the pre-registered reference cell across five training seeds on the same corrected partition gives corrected accuracies 0.8028–0.8131 (mean 0.8086, sd 0.0040) against the linear SVM's 0.7975, and as-shipped accuracies 0.8119–0.8359, between the forest's 0.8596 and the linear SVM's 0.7980, so the rank is the same on every seed. The exchanging pair is the MLP and the CNN, and both arms of the inversion criterion clear zero: as-shipped MLP$-$CNN $+0.0500$ $[0.0320, 0.0679]$, corrected CNN$-$MLP $+0.0402$ $[0.0309, 0.0495]$. Against the harder comparator the criterion does not require, *LinearSVC*, the paired difference is $+0.0111$ with a combined-source interval of $[0.0019, 0.0204]$, folding the seed-to-seed standard deviation (0.0040) into the cluster-bootstrap standard deviation (0.0026) in quadrature, because the quantity of interest is the advantage of *one* trained network rather than of an average over five. Bootstrapping seed-averaged per-example correctness instead gives $[0.0063, 0.0161]$, narrower than any individual seed's interval because averaging removes the training variance rather than estimating it, and we do not quote that as the result. Four of the five individual seeds have paired intervals excluding zero and the fifth does not ($+0.0053$, $[-0.0006, 0.0109]$), so the advantage is uniform in rank but not in significance. The reference cell is the *maximum* of the nine-cell regularization grid, whose corrected accuracies span 0.749–0.813 with 0.7975 inside—five cells above, four below—so no conclusion is drawn from that cell alone; the claim rests on seed replication of one pre-specified cell against a paired interval.

human_nontata_promoters: a dead heat.

On that dataset the picture is weaker. Under accuracy the forest is demoted from rank 1 to 3 (corrected LR 0.829 $>$ HGB 0.827 $>$ RF 0.820 $>$ LinearSVC 0.808, all seed-0; the forest's 0.820 is the seed-0 value of the quantity whose five-seed mean, 0.810, gives the headline drop). This is a statistical dead heat—corrected CIs overlap (RF $[0.812, 0.828]$, LR $[0.821, 0.837]$)—and does not survive a threshold-free metric: under AUROC and F_1 the forest remains rank 1 (corrected AUROC 0.929, highest of the four). With only four models Kendall τ is too coarse for a primary statistic. The nine-model roster produces a larger scramble there with supported swap margins, but the prevalence-corrected graded gap does not separate the model families on it (memorizers $+0.122$ against $+0.179$), so the wider roster does not strengthen that dataset's case; an alternative discriminator fails too, the residual of each drop against φg being *larger* on *human_enhancers_ensembl* ($+0.257$) than on *human_nontata_promoters* ($+0.152$), so it measures the retraining penalty rather than dataset quality.

The reordering therefore stands as an existence proof on *human_enhancers_ensembl* and as a cautionary partial case on *human_nontata_promoters*.

The six registered predictions, scored. P1 fails as registered: on the raw gap the largest value belongs to the forest on *human_enhancers_ensembl* (0.230 against 1-NN's 0.208 and extremely randomized trees' 0.210) and to 15-nearest-neighbour on *human_nontata_promoters*. Under the prevalence-corrected gap 1-NN does lead on the first ($+0.461$ against $+0.319$), but that statistic was defined after P1 was committed and the reading is *post hoc*, not a confirmation. P2 holds on *human_enhancers_ensembl* (1-NN has the largest drop, 0.336) and fails on *human_nontata_promoters*, where both P1 and P2 fail. P3 holds decisively on the first (mean rank 4.50 against 5.50 as shipped, 7.25 against 1.50 corrected). P4 holds: on the clean control no model's drop interval excludes zero, for any of the nine—that control runs on a $20{,}000$-sequence subsample of *human_enhancers_cohn* rather than its full $27{,}791$, admissible because its full-scale leak fractions are 0.001 and 0.005. P5 and P6 are scored in §3.10 and Supplementary §S2.

3.7. The mechanism (exploratory), and its scope

On *human_enhancers_ensembl* the near-default forest scores 1.000 on near-duplicate test sequences (≥ 0.9 similarity) but 0.770 on novel ones—a raw graded gap of $+0.230$ against 0.034–0.037 for the other three learners of the four-model comparison, though across the wider roster the raw gaps reach 0.210 and 0.208. On novel sequences the forest (0.770) is in fact *worse* than the linear SVM (0.782). Those are the registered quantities, and on them P1 fails.

Exploratory: the prevalence-corrected gap.

Everything from here to the end of this subsection is exploratory and is reported as such. P1 was committed against the *raw* graded gap, and the prevalence-corrected form below was defined after that prediction had been scored, so it cannot confirm the hypothesis it was built to rescue—it can only motivate the prospective test we register at the end of this paragraph. Correcting the class-composition confound *strengthens* the mechanism: from balanced accuracy, 1-nearest-neighbour, the definitional memorizer, has by far the largest gap ($+0.461$), then extremely randomized trees ($+0.370$) and the forest ($+0.319$), with the linear models ($+0.154$, $+0.144$), naive Bayes ($+0.129$) and, lowest, 15-nearest-neighbour ($+0.016$)—averaging over fifteen neighbours is what stops a nearest-neighbour rule memorizing. The four memorization-prone learners average $+0.292$ against $+0.165$, a separation the raw statistic does not produce. Memorization pays because near-duplicates carry the label: at ≥ 0.9 similarity 99.99% of near-duplicate test sequences share their nearest training neighbour's label on *human_enhancers_ensembl* ($n = 11{,}774$) and 99.95% on *human_nontata_promoters* ($n = 2{,}037$); clean datasets have near-empty high-similarity bins (Supplementary Fig. S2). The correction is equally informative where it removes a signal: on *human_nontata_promoters* the corrected gaps do *not* separate the families—memorizers average $+0.122$ against $+0.179$ for the other five, the largest belonging to the MLP—so the mechanism claim is scoped to *human_enhancers_ensembl*. We therefore register the corrected gap forward rather than counting it backward: on the next leaky dataset audited, 1-nearest-neighbour will carry the largest prevalence-corrected graded gap and the four memorization-prone learners will average a larger corrected gap than the remaining five. That prediction is falsifiable on one dataset, and *human_nontata_promoters* above is the case in which it would already have failed.

Exploratory: the shape of the curve, against a published one.

That accuracy grades with training-set similarity, and that models rely on memorized associations where it does, is established by Rafi et al. (2026) across three modelling regimes; it is not claimed as novel here, and what this suite adds is a matched clean control and an editable construction step. Their curve is non-monotonic—best at both extremes of homology, worse at intermediate similarity—so ours is worth reporting whether or not it agrees. Binning test sequences by maximum similarity to training and reading *balanced* accuracy, so the prevalence confound above cannot manufacture a shape, the two suites disagree in an interpretable way. The memorizing forest is monotone increasing on both leaky datasets ($0.681 \rightarrow 0.818 \rightarrow 0.910 \rightarrow 1.000$ on *human_enhancers_ensembl*; $0.788 \rightarrow 0.922 \rightarrow 0.997 \rightarrow 1.000$ on *human_nontata_promoters*), which is what a memorization account predicts and what their non-monotonic curve does not show. The models that do *not* memorize are non-monotonic on both, but in opposite directions: on *human_nontata_promoters* logistic regression dips at intermediate similarity ($0.824 \rightarrow 0.755 \rightarrow 0.800 \rightarrow 0.963$), the shape they report, while on *human_enhancers_ensembl* it peaks there instead ($0.747 \rightarrow 0.955 \rightarrow 0.886 \rightarrow 0.901$). Two things limit this. The intermediate bins are small on *human_enhancers_ensembl*—311 and 115 sequences against $18{,}770$ and $11{,}774$ at the extremes—so the peak rests on few observations; and their regimes reach 196 kb contexts and functional divergence between homologs, neither of which a k-mer model on ≤ 573 bp windows can express. We report the difference rather than resolve it (`results/graded_shape_summary.csv`).

3.8. The effect is a property of the unregularized forest

At `min_samples_leaf=1` the forest scores exactly 1.000 on the ≥ 0.9 bin, with drops of 0.164 and 0.108 (seed 0), graded gaps 0.230 and 0.121, and rank 1. At 500 the drop is ≈ 0 ($0.164 \rightarrow -0.001$; $0.108 \rightarrow 0.000$) and the leaf-size gap collapses ($0.230 \rightarrow 0.020$; $0.121 \rightarrow 0.061$), monotonically but for one 0.0012 reversal in the last *human_nontata_promoters* step; along the depth axis the gap is non-monotone there, rising to 0.224 at depth 16 before falling to 0.161. Crucially, the "forest wins on the leaky split" phenomenon itself *requires* the unregularized forest: as shipped it holds rank 1 only near the default and has fallen to rank 4 by `min_samples_leaf`$\geq$ 20 on both datasets. Regularization removes the inflation without recovering the accuracy the inflation concealed—corrected accuracy on *human_nontata_promoters* falls $0.820 \rightarrow 0.785$ along the same path—so the two interventions are not interchangeable.

3.9. Deep models, binary and multiclass

A from-scratch 1D CNN trained only on each training split shows the same inflation, so the effect is not a classical-model artifact. The pre-registered reference cell drops on *both* datasets—$+0.043$ on *human_nontata_promoters* $[0.025, 0.061]$, $+0.014$ on *human_enhancers_ensembl* $[0.007, 0.021]$—so the refutation condition, which requires the interval to include zero on *both*, is not triggered. One *human_enhancers_ensembl* seed does include zero, and *human_nontata_promoters*, which carries the larger drop, was not seed-replicated and is the more dispersed of the two across the grid, so the verdict rests on an unreplicated cell on the dataset from which §3.7 scopes the mechanism away. Nineteen of 20 cells drop with intervals excluding zero; the exception is the *human_nontata_promoters* dropout-0.6 cell, where accuracy *rises* 0.020. Seed replication of the *human_enhancers_ensembl* reference cell gives drops of 0.0049–0.0269, a five-fold spread whose smallest does not exclude zero, so $+0.014$ is one draw and no single-run drop here, including each of the nineteen, is a point estimate. One registered expectation was not met: the pre-registration named this a *dose-response* grid, and it is not one, the drop being non-monotone in both dropout and weight decay.

The multiclass question is answered by construction, and the two model families answer it differently. On the identical injected construction the forest's drop grows monotonically with f: $+0.0004$, $+0.1177$, $+0.1791$, $+0.2581$ at $f = 0$, 0.1, 0.2, 0.4. Logistic regression is not untouched—it drops 0.0158, 0.0323, 0.0661, monotone in f, its as-shipped accuracy climbing $0.591 \rightarrow 0.669$ as it exploits the injected copies—but the forest's advantage is $7.4\times$ to $3.9\times$ larger, narrowing as the dose rises. The CNN shows *no* dose-response: mean drops -0.0386 (sd 0.0239), -0.0182 (0.0296), -0.0249 (0.0325), -0.0089 (0.0362) over three seeds at the same four doses (each cell a single run)—every mean negative, the seed standard deviation equalling or exceeding any dose effect. All three seeds are reported; seed 1 underfits, its as-shipped accuracy averaging 0.6856 against 0.7552 and 0.7534, and is retained rather than excluded. The mechanism therefore generalizes to multiclass *for the classical memorizer* and not for the regularized network—what "memorization propensity, not capacity" predicts, since the label-carrying copies are available to every learner and the differential is a property of which learner exploits them.

3.10. Ordinary tuning selects the memorizing configuration

A practitioner does not know a split is leaky; they tune by cross-validating on the training set. On *human_enhancers_ensembl*, ordinary stratified 5-fold selects `min_samples_leaf`$= 1$, the memorizing default, and the model it selects scores 0.860 as shipped but 0.696 corrected—a loss of 16.4 points. Cluster-grouped 5-fold selects 20, and that model loses only 2.2 points ($0.751 \rightarrow 0.729$). Worse, the benchmark punishes the correct methodology: the leakage-aware tuner's model appears 10.9 points *worse* as shipped (0.751 against 0.860) while being 3.3 points *better* corrected (0.729 against 0.696)—a point-estimate comparison to which we attach no interval. On *human_nontata_promoters* both procedures select 1, so P5 holds on the first dataset and is a tie, not a reversal, on the second. The untuned default is therefore not a careless choice a tuner would correct; it is what ordinary model selection selects, and the benchmark rewards it.

3.11. A second suite carries the same defect: three of eleven independent tasks are leaky

Counting independent tasks, three of eleven Nucleotide Transformer tasks are leaky and six more borderline, leaving two clean. *enhancers* carries 25.0% test-to-train leakage that is *byte-identical*—verified on a code path using no k-mers, 100 of 400 test sequences at label concordance 1.000—while *splice_sites_acceptors* and *_donors* carry 0.244 and 0.245 at Jaccard 0.7 with zero exact duplicates. An earlier pass capped training sets at $20{,}000$ and returned three borderline and five clean; re-running uncapped moved three histone tasks across the cut on containment.

The splice mechanism is not the cross-species homology an earlier account proposed: 65.9% and 68.3% of leaked test sequences share the same UniProt entry name with their best training partner—same gene, same organism—and at Jaccard ≥ 0.9 that share *rises* to 77.1% and 82.6%, where homology predicts thinning. It is same-gene splice-window redundancy, and label concordance among leaked pairs is correspondingly poor (0.819, 0.792 against 0.999 on both Genomic Benchmarks datasets). That account is offered after the fact, not predicted (Supplementary §S6).

Running the full pipeline there returns a null: no material ranking inversion on any leaky task nor on the control. Leakage does inflate scores on *enhancers*—the forest drops 0.0413 $[0.0241, 0.0583]$, *LinearSVC* 0.0221 $[0.0055, 0.0381]$, HistGradientBoosting 0.0185 $[0.0021, 0.0366]$, three of the four learners excluding zero, while logistic regression, leading on both splits, is null ($+0.0046$ $[-0.0104, 0.0199]$). The memorization *ordering* reproduces—the forest carries the largest drop and is the only learner with a positive graded gap ($+0.100$ against -0.084 to -0.169)—but the confinement to memorizers reported on *human_enhancers_ensembl* does not: two non-memorizers exclude zero here, and their negative graded gaps rule out memorization, leaving the retraining penalty of §3.6 the better reading. On *splice_sites_acceptors* the memorizing forest's drop is null ($+0.0201$ $[-0.0068, 0.0446]$), so the model carrying the mechanism is the one the effect does not reach. On *splice_sites_donors* correction *raises* accuracy for three of four learners and one gain is nominally supported (*LinearSVC* -0.0251 $[-0.0526, -0.0004]$), marginally and seed-dependently ($p \approx 0.059$ uncorrected). Under Benjamini–Hochberg across this cross-suite grid only *enhancers*|RF survives, so these flags are nominal. The leading model is unchanged on all four tasks, which is the claim this section rests on: reordering additionally requires a challenger genuinely better on novel sequences, and this suite supplies leakage without one.

3.12. A pre-registered cross-suite test, falsified

Our pre-registration carried binding predictions for GUE (Zhou et al., 2023b), recorded before that suite was examined—the prediction document was committed as `2edad5c` on 2026-07-18 and the census that scored it as `3af3eb4` nine hours later, in separate commits whose ordering is published with the `git log` output in `PREREGISTRATION.md`, subject to the limits of author-controlled timestamps stated there: that its short fixed-length human regulatory tasks—core promoter at 70 bp, promoter at 300 bp, transcription-factor binding at 101 bp—would be leaky, and its multi-species tasks clean. At full scale the registered predictions score 3 of 15, and all eleven predicted-leaky tasks are clean, measuring 0.009 to 0.041 under Jaccard and 0.014 to 0.042 under containment—little leakage, not none, the largest under half the verdict cut. These are not eleven independent benchmarks: the *_all* promoter tasks' test sets are the exact union of their own *notata* and *tata* test sets, and the 70 and 300 bp promoter families are the same loci at two window widths, leaving seven independent test partitions, which bounds how much independent evidence the null carries.

This refutes the hypothesis the prediction encoded. Leakage does not track task type: seven independent benchmarks in precisely the regime we flagged do not carry it. What it tracks is the construction step of §3.2—Genomic Benchmarks' two leaky datasets have an un-deduplicated assembly where GUE's comparable tasks do not. Two verdicts moved. *emp_H3K4me3*, registered clean, has a full-scale containment leak fraction of 0.138 against 0.093 capped, making it borderline: under the binary rule committed in advance it remains a correct prediction and the score stands at 3 of 15; under a three-way rule it is a miss and the score is 2 of 15. And *virus_covid*, also registered clean, measures 1.000—the twelfth miss, and the one discussed in §4, where it is not a curation defect. Overall, the prediction that leakage follows task type is refuted by seven independent partitions in precisely the regime it named; the two verdicts that moved are borderline calls, not reversals.

3.13. Robustness

The block bootstrap.

That the intervals move little is not because the dependence is absent. The splitter assigns whole components to one side, so every near-duplicate relation surviving correction is test-to-test by construction, and the corrected test sets are the more clustered arm: 43.4% of corrected *human_nontata_promoters* test sequences lie in a multi-member component against 25.2% as shipped, and 47.7% of *human_enhancers_ensembl*'s against 10.1%. Those sizes are heavy-tailed, so Kish's m^* is 4.45 and 2.58 where the mean is 1.55 and 1.33, giving design effects of 2.80 and 2.58 for the forest where the mean-size convention returns 1.29 and 1.33. The inflation is measured directly: on the corrected arm the cluster-bootstrap variance of the forest's accuracy exceeds its sample-bootstrap variance by $3.96\times$ and $2.12\times$ ($3.10\times$ and $1.78\times$ for logistic regression), widening the delta intervals 1.2–$1.8\times$. The intraclass correlation is high (0.78–0.99) across the four fits, so the design effect is governed by the block-size distribution rather than by the within-block correlation. The dependence is therefore real, larger than a mean-size design effect suggests, concentrated in the arm the correction creates—and already inside every reported interval, because the block bootstrap resamples those components rather than modelling them. No interval and no verdict changes: the forest's drop still excludes zero on both datasets ($[0.155, 0.173]$; $[0.092, 0.126]$), and Benjamini–Hochberg over the declared 15-delta family leaves the same set significant. One source no test-set resampling reaches is the fit, which is held fixed.

Chromosome holdout.

The demotion reproduces on *human_enhancers_ensembl*: the forest falls from rank 1 (0.8596) to rank 4 (0.7069), giving the identical corrected order as the near-duplicate-aware split. On *human_nontata_promoters* it does *not*: the forest stays rank 1 ($0.9284 \rightarrow 0.8199$), a third independent signal—with the AUROC/F_1 reversal and the novel-only rank correlation—that its demotion is accuracy-only. Two disclosures bound that arm: its test set is 74.33% positive against a 47.20% training prevalence, a 27-point shift, and it is scored on accuracy, which the imbalance panel below shows understates the correction. The controls agree on the drop magnitude on *human_enhancers_ensembl* (corrected 0.707 against 0.696); on *human_nontata_promoters* they agree to 0.0001 across test sets 74.3% and 54.4% positive, which is consistent with, but no evidence for, the near-duplicate-aware split proxying the deployment control.

Metric geometry and cluster structure.

The demotion survives canonical (reverse-complement-collapsed) k-mer features (forest $1 \rightarrow 4$, drop 0.153 on *human_enhancers_ensembl*; $1 \rightarrow 3$, drop 0.097 on *human_nontata_promoters*), so it is not forward-strand overfitting, and a GC-in-distribution restriction gives *lower* corrected accuracy (0.787, 0.631), not the inflated original, so it is not a GC covariate shift. An MMseqs2 alignment-scored re-clustering reproduces the effect on *human_enhancers_ensembl* (0.164 against 0.162) and gives a milder drop on *human_nontata_promoters* ($0.108 \rightarrow 0.064$); that arm changes clustering engine and linkage rule together, so the difference is a linkage signature as much as a metric one (Supplementary §S3). Cluster cohesion explains the asymmetry between the two datasets, and geometrically rather than biologically: *human_enhancers_ensembl*'s clusters are 99.6% pairwise at cohesion 0.96, discrete $2\times$ locus duplication that whole-cluster correction removes cleanly, whereas *human_nontata_promoters*'s 420-member largest component sits at cohesion 0.083, is $420/420$ *negative* class, and is a single locus tiled by 251 bp windows at a median 2 bp step—window geometry, not chained paralogy. Dataset-wide 94.3% of its negatives sit in multi-member components against 0.4% of positives and $2{,}808$ of $2{,}811$ components are class-pure, the same fact §3.2 reports as $R_{\mathrm{self}} = 0.961$ in the negative class; the certification tool therefore returns *leaky-but-correction-unsafe* rather than *leaky* below cohesion 0.5. A `dustmasker` check excludes a microsatellite artifact but is silent on interspersed repeats, and on *human_nontata_promoters* the leaked stratum is in fact $\sim 7\times$ AluY-enriched relative to novel (Supplementary §S3).

Imbalance, and what "corrected" means.

Under induced prevalence imbalance the inflation is revealed by prevalence-aware metrics and masked by accuracy: on *human_enhancers_ensembl* at $\pi = 0.2$ correction lowers AUPRC $0.622 \rightarrow 0.477$, MCC $0.422 \rightarrow 0.182$ and minority recall $0.258 \rightarrow 0.070$ while accuracy barely moves ($0.844 \rightarrow 0.808$), and on *human_nontata_promoters* AUPRC falls $0.949 \rightarrow 0.813$

and MCC $0.785 \to 0.682$ against accuracy's $0.934 \to 0.903$—but minority recall is flat there ($0.673 \to 0.674$), so the recall collapse does not replicate. What generalizes is that AUPRC and MCC fall several times further than accuracy on both datasets, so accuracy at threshold 0.5 understates the correction; the panel is random-forest-only and its prevalence induced rather than native, leaving the reordering claim untested under imbalance. Across the four robustness arms—block bootstrap, chromosome holdout, metric geometry and induced imbalance—no interval and no verdict changes, and the dependence the block bootstrap measures is real, larger than a mean-size design effect suggests, and already inside every reported interval.

4. Discussion

Near-duplicate leakage in this suite is produced by a step in how the data were assembled, and that step is visible, editable and reversible before any model is trained: a coordinate signature carrying no sampling uncertainty and no model dependence, a manipulation that removes and imposes the defect at will, and a pre-registered task-type prediction that failed on all eleven targets. On one dataset the downstream cost is that the benchmark reports the wrong winner.

The reordering requires the *unregularized* forest, which by `min_samples_leaf`≥ 20 has already fallen to rank 4 as shipped (§3.8). That configuration is not a careless choice a practitioner would correct: ordinary stratified cross-validation on the as-shipped training set selects it, and the benchmark then penalises the practitioner who tunes correctly, showing the leakage-aware tuner's model 10.9 points worse as shipped and 3.3 points better corrected (§3.10). The property that matters in a benchmark is anyway not whether some user tunes but whether the data reward memorization at all, and §3.3 locates that reward in the construction step. The controlled variable is memorization propensity, not capacity: the nominally higher-capacity HistGradientBoosting is barely inflated where the near-default forest is heavily inflated, and the regularized CNN shows no dose-response under injection at all—its mean drop is negative at every dose—where the near-default forest's grows monotonically to $+0.258$ (§3.9).

The suite's own published baseline is a convolutional network (Grešová et al., 2023), not a tree ensemble. A CNN-based leaderboard is affected in the direction that costs the architecture: our from-scratch network ranks fifth of ten as shipped and first corrected, scoped to one dataset and to a *post hoc* comparison, with one of its five seeds not individually clearing zero (§3.6). Its inflation is between six- and thirtyfold smaller than the forest's depending on seed, so the ratio is seed-dependent and is reported as a range; the reordering condition is not met by a CNN-only comparison, and the result is bounded by the comparison set: ten learners, one dataset.

One interpretive point governs every corrected number in this paper. A near-duplicate-aware split answers how a model scores on test sequences with no near-copy in training, which is not how a deployed model will perform, since paralogues, gene families and conserved regulatory elements are real inputs it will meet, and whole-cluster assignment also strips near-duplicate partners from the *training* set (§3.6). "Corrected" therefore qualifies the split and not the accuracy: a corrected accuracy is a lower bound on deployment performance under one adversarial notion of novelty. The as-shipped ranking is wrong; the corrected accuracy is a bound, not a deployment estimate.

human_enhancers_ensembl is an existence proof: the ranking change holds under accuracy, AUROC, F_1, a bootstrap winner probability, a chromosome-holdout control and an alignment-scored re-clustering. *human_nontata_promoters* is a cautionary partial case, five independent signals showing its demotion is accuracy-only: overlapping corrected CIs, the AUROC/F_1 reversal, the chromosome control, the novel-only rank correlation and the milder alignment drop. Its clusters are the geometric case rather than the duplication case, so whole-cluster correction is a coarser instrument there. Because the effect is dataset-specific the contribution is constructive: not that these benchmarks are broken, but which datasets to trust and a tool to certify a new one. Green verdicts carry a heavier evidential burden than red ones, and Table 1 states what a "clean" entry does and does not certify.

Scale matters practically. Subsampling can mask leakage—*human_enhancers_ensembl* appears almost clean at 20k sequences and is heavily leaky at full scale—so audits must run at full dataset scale. Both censuses reported here were capped at $20{,}000$ training sequences on a first pass and were wrong the same way: re-run in full, GUE's leak fractions rose by up to $2.8\times$ on the tasks large enough to have been capped and one verdict moved clean to borderline, and three Nucleotide Transformer histone tasks crossed the cut, leaving three of eleven independent tasks leaky and six borderline (§3.11, §3.12).

The evidence spans more than regulatory-element classification (Fig. 4): the fifty-one dataset–task censuses cover six task types and six source organisms, and forty-four of the fifty-one fall left of the verdict cut, including all eleven GUE tasks registered in advance as leaky. Length is the weaker axis: no censused task exceeds a $1{,}000$ bp median in sequence space, and the one long-range datapoint is in coordinate space only (§3.4). Protein sequence is out of scope for this detector: an 8-mer over a twenty-letter alphabet collapses the Jaccard towards exact match, so a protein census would need k re-chosen and the calibration redone—a different instrument.

What the detector detects, and one false positive.

The 8-mer Jaccard at 0.7 is not a fixed instrument across length (§2.4), so every leak fraction is a lower bound whose tightness depends on length. Specificity is measured rather than argued. Adjudicating a band-stratified sample of test–train pairs by Smith–Waterman against the CD-HIT-EST-like criterion the curators omitted ($\geq 50\%$ coverage of the shorter sequence at $\geq 80\%$ identity), every flagged band on both leaky datasets is 100% homologous: precision 1.000 (Clopper–Pearson lower bounds 0.951 and 0.978) and a per-pair false-positive rate of 0.000, bounded above by 1.5×10^{-7} and 1.1×10^{-6}—well below the $1/n_{\text{train}}$ scale at which a maximum-over-training-set statistic would be corrupted. Recall runs the other way, 0.006 and 0.260, because homology at 80% identity over half a sequence routinely leaves the 8-mer Jaccard below 0.7; that is the quantitative form of the lower-bound scope above, not a defect (Supplementary §S4). Construction and provenance now corroborate the measurement rather than substitute for it: the merged arm of fix-a-leaky is clean as a property of the edit itself and reads 0.012, the dose-zero control 0.004, and the five clean Genomic Benchmarks datasets, whose merge-and-deduplicate

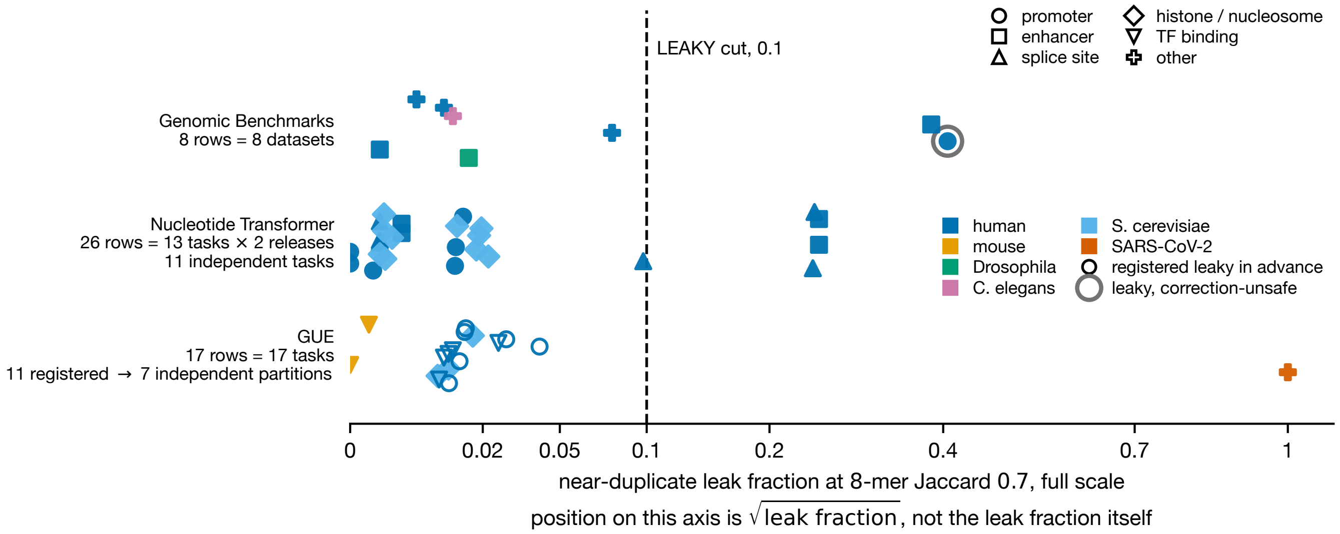

Figure 4 Near-duplicate leak fraction across the fifty-one dataset–task censuses reported here, in three benchmark suites. Each point is one censused dataset–task *row*, which is not the same as an independent task, and the panel annotates both conventions: the Nucleotide Transformer contributes 13 shipped tasks in two releases (26 rows) that collapse to 11 independent tasks, GUE ships 17 tasks of which the 11 registered in advance collapse to 7 independent test partitions, and the text reports independent-task counts throughout. x is the fraction of test sequences whose maximum 8-mer Jaccard to any training sequence exceeds 0.7, measured at full scale; position on the axis is the *square root* of that fraction, as the axis label states, so the sub-0.1 region, which holds forty-four of the fifty-one, is legible. Marker shape denotes task type (13 promoter, 7 enhancer, 6 splice site, 13 histone-mark or nucleosome occupancy, 7 transcription-factor binding, and five singletons: open chromatin, coding-versus-intergenomic, human-versus-worm, three-class regulatory assignment and nine-way viral variant assignment); colour denotes source organism. The dashed rule at 0.1 is the LEAKY verdict cut used throughout. Open symbols are the eleven GUE tasks registered in advance as leaky; all eleven fall left of the cut, at 0.009 to 0.041. The single grey-haloed point is *human_nontata_promoters*, the one dataset certified *leaky-but-correction-unsafe* (Table 1). The SARS-CoV-2 point at 1.000 is *virus_covid*, where near-identity is a property of the genome rather than a curation defect (§4). A fourth suite, BEND, is censused in coordinate space only and appears in Fig. 1B.

provenance is documented, 0.001 to 0.016, against the two leaky datasets' 0.384 and 0.406. The GUE tasks cannot join that list: their "clean" label *is* this estimator's output, so quoting it would be circular. What they supply independently is the exact-duplicate rate, by string equality with no k-mers, running 0.000 to 0.026 across the eleven and agreeing with the k-mer verdicts through a separate code path.

The detector also cannot distinguish a curation defect from biology. GUE's *virus_covid*, registered clean, measures a near-duplicate fraction of 1.000, the largest we observed anywhere, and an exact-duplicate rate of 0.122—yet it is not a curation defect. The task is nine-way SARS-CoV-2 variant classification over 999 bp windows of a ~ 30 kb genome whose variants differ by a handful of mutations, so test-to-train similarity has a *minimum* of 0.824 and a median of 0.987: no split could be otherwise, and a leak fraction carries no information there. Any application to viral, organellar or highly conserved sets must establish the distinction by other means—and the coordinate remedy is unavailable on viral releases, which ship no coordinates.

Pretrained models and the second leakage channel.
Those models are contaminated by construction, so fine-tuning them here would produce a confounded number rather than evidence: the assembly is verified GRCh38 (§2.2). The overlap is nonetheless not total, and the exception is useful. HyenaDNA inherits the Enformer interval set (Avsec et al., 2021) and designates chromosomes 14 and X as its held-out pretraining test chromosomes; both are present here, carrying 6.6% of *human_enhancers_ensembl*'s test set and 15.3% of *human_nontata_promoters*'s, the latter unevenly by class (24.6% of test negatives against 7.5% of positives). Pretraining overlap is therefore at most $\sim 93.4\%$ and $\sim 84.7\%$ for that model, not $\approx 100\%$; for DNABERT-2 and the Nucleotide Transformer, whose human component is one part of a multispecies corpus, the argument remains one from provenance.

One logical point cuts in our favour: pretraining is label-free and common to both split arms, so it *attenuates* the as-shipped-minus-corrected difference rather than inventing it, and a positive finding under contamination is conservative evidence. What would settle the question is an overlap-free arm, and chromosomes 14 and X are a stratum HyenaDNA provably never saw—the right future protocol, and one these datasets already contain enough of to carry an interval. A second route, priced but not performed, applies break-a-clean to *drosophila_enhancers_stark* and fine-tunes HyenaDNA on both arms under both splits, manufacturing the contamination inside a genome the model has never seen: $6{,}914$ sequences of at most $3{,}237$ bp and twelve runs of `hyenadna-small-32k`. Whether deployed foundation models inherit this inflation on human benchmarks is left open; our from-scratch CNN establishes only that the mechanism reaches a trained neural network with no pretraining confound.

The obvious next targets—DeepSTARR (de Almeida et al., 2022), BEND (Marin et al., 2024) and DART-Eval (Patel et al., 2024)—sharpen our prediction rather than confirm it.

Each assembles its examples from genome-tiled or annotation-overlapping intervals, the construction step that produced the defect here, but each also partitions by genomic position or sequence identity: DeepSTARR bins STARR-seq peaks into 249 bp windows at a 100 bp stride yet holds out the two halves of chromosome 2R; DART-Eval holds out chromosomes 5, 10, 14, 18, 20 and 22; BEND partitions by chromosome on three tasks and at 80% identity on the fourth. Overlapping intervals therefore cannot span those splits, and our condition predicts all three clean—confirmed directly for BEND in §3.4. The condition our evidence supports is thus conjunctive: overlapping construction *and* a split that ignores genomic position—the pair this paper's title names, and the reason neither the construction step nor the split rule is separately diagnostic. Our own suite makes the same point, since *drosophila_enhancers_stark* tiles at a $2{,}142$ bp median and audits clean because its cross-split overlaps are base-pair-scale touches rather than shared windows.

5. Conclusion

Near-duplicate leakage in a widely used genomic benchmark suite is a curation defect with a signature in the shipped coordinates, and it does more than inflate scores: on one dataset it changes which model the benchmark reports as best. Two of the seven binary datasets are leaky—one of them only in the weaker *correction-unsafe* sense—one borderline under a length-robust metric and four clean; the cause is $77{,}421$ positive intervals on $40{,}934$ distinct coordinates (§3.2), and editing that construction step alone both abolishes the syndrome and manufactures it in a clean dataset (§3.3). Correcting the split costs the best model 16.5 points on *human_enhancers_ensembl*, combined-source $[15.4, 17.4]$, and 10.8 points on *human_nontata_promoters*, $[8.8, 12.9]$—each interval quoted on the estimand it was computed on, the latter the seed-0 partition whose five-seed mean is the 11.8 points §3.5 reports—both excluding zero where every clean delta includes it, and on *human_enhancers_ensembl* a nine-learner roster cannot separate its top three as shipped yet those three span 23 accuracy points corrected (§3.6). *human_nontata_promoters*, where the demotion is accuracy-only, is a cautionary partial case rather than a second demonstration.

Four practices follow. Audit at full dataset scale, since subsampling can make a heavily leaky dataset look clean—a subsample is admissible to confirm cleanliness, never to establish it. Report both a length-blind and a length-robust similarity, since containment flagged as borderline one dataset the Jaccard scored clean, and let the threshold follow sequence length. Partition in coordinate space before splitting, since the defect here was introduced upstream of any sequence-level analysis—though coordinates may escalate a verdict and never clear one, and an exact-coordinate collapse is a no-op on datasets like *human_nontata_promoters*, whose redundancy is in overlapping windows rather than repeated rows. And include at least one memorization-prone learner in any comparison, since the gap between its leaky and corrected scores is a direct tripwire for a contaminated split, costlier than the as-shipped screens because it requires building the corrected split. All four transfer unchanged to multiclass, where per-class prevalence should additionally be checked after splitting.

The reordering result is a single-dataset existence proof, scoped to comparisons containing a memorization-prone model. What generalizes is the cause: near-duplicate leakage here is produced by an un-deduplicated assembly step, is visible in shipped coordinates before any model is trained, can be removed and imposed at will with the downstream consequences following in both directions, and is absent from the eleven short human regulatory tasks—seven independent partitions—that a task-type hypothesis predicted would carry it. Audit the construction step and the split rule together, not the sequence type.

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Conflict of Interest

None declared.

Data Availability

All code required to reproduce the analyses, along with the report card, figures, the near-duplicate-aware splitter and the certification tool, is available at `https://github.com/rikhinkavuru/homology-leakage-genomic-benchmarks`.
The Genomic Benchmarks datasets analysed here are publicly available and were obtained via the genomic-benchmarks package (Grešová et al., 2023); they are not redistributed in our repository. The coordinate-space census of BEND (§3.4) uses only the `bed` annotation files distributed with that resource (Marin et al., 2024), which carry their own split assignments and require no reference genome; those files are likewise not redistributed, and the script that downloads them and reproduces the census is included. Every upstream reference-genome and annotation URL on which the datasets depend—Ensembl releases 97 and 100, and the UCSC `hg38/hg19` REST endpoints—was re-resolved on 2026-07-20 and returned HTTP 200; the release, artifact, live URL, archive host and access date for each are tabulated in `results/reference_provenance.csv`. Because Ensembl release 116 (June 2026) is the final release served on the current platform, we cite the dated archive hosts (`https://jul2019.archive.ensembl.org`, `https://apr2020.archive.ensembl.org`) alongside the live paths.

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